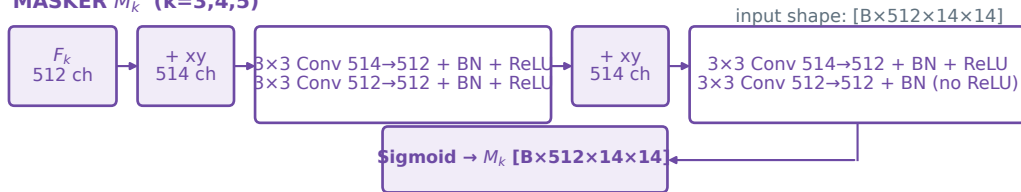


D Causal feature partition

MASKER M_k ($k=3,4,5$)

three Maskers: same structure, separate parameters



COMPLEMENTARY SPLIT AND SHARED AGGREGATION



E Two independent projectors

VISUAL DECODING — one active ATConv in each head

supportive P_{sup} and inferior P_{ad}



TEXT-CONDITIONED DYNAMIC CLASSIFIER

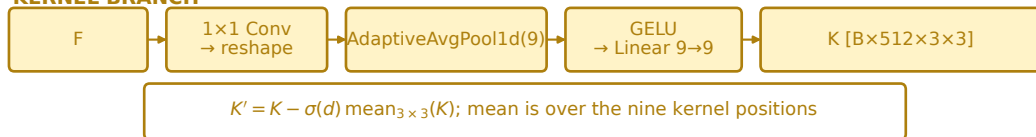
$s_p [B \times 1024] \rightarrow$ Linear $1024 \rightarrow 2305 \rightarrow$ split into $W [B \times 256 \times 3 \times 3]$ and $b [B]$
 $\rightarrow$ per-sample grouped dynamic Conv 3×3 over $U \rightarrow$ branch logit $[B \times 1 \times 224 \times 224]$

$z = (z_{sup} + z_{ad})/2$ | train: $0.5 \text{ BCEWithLogits}(z, y') + 0.5 L_{Dice}(\sigma(z), y')$ | test: $P = \sigma(z), \hat{y} = 1[P \geq 0.5]$

F ATConv internal operator

visual-only; exactly two effective instances

KERNEL BRANCH



FEATURE BRANCH

$F \rightarrow 1 \times 1$ Conv $\rightarrow$ reshape $[1, B \times 512, H, W] \rightarrow$ dynamic depthwise grouped Conv using K' groups= $B \times 512 \rightarrow$ reshape $\rightarrow 1 \times 1$ Conv $\rightarrow$ ATConv output $[B \times 512 \times H \times W]$

s_p does not enter ATConv